

GCL BENTOMAT® CL

REINFORCED GEOSYNTHETIC CLAY LINER (GCL)

PROPERTIES	TEST METHOD	GCL BENTOMAT® – BCL
		IMPERIAL
² BENTONITE MOISTURE CONTENT	ASTM D2216	12% max
² BENTONITE SWELL INDEX	ASTM D5890	24 mL/2g min
² BENTONITE FLUID LOSS	ASTM D5891	18 mL max
³ BENTONITE MASS/AREA	ASTM D5993	0.75 lb/ft ² (3.7 kg/m ²) min
³ TOTAL MASS/AREA	ASTM D5993	0.84 lb/ft ² (4.1 kg/m ²) min
¹ GEOFILM DENSITY	ASTM D1505	0.92 g/cm ³
¹ GEOFILM THICKNESS	ASTM D5199	5 mil (0.12 mm) min
^{1,4} GEOFILM BREAK STRENGTH	ASTM D882	14 lbs/in (2.5 kN/m) min
GCL MOISTURE CONTENT	ASTM D5993	35% max
⁵ GCL GRAB STRENGTH	ASTM D6768	30 lbs/in (5.3 kN/m) min
GCL PEEL STRENGTH	ASTM D6496	3.5 lbs/in (610 N/m) min
⁶ GCL HYDRAULIC CONDUCTIVITY	ASTM D5887	5 x 10 ⁻¹² m/s max
⁶ GCL INDEX FLUX	ASTM D5887	1 x 10 ⁻⁹ m ³ /m ² /s max
⁷ GCL HYDRATED INTERNAL SHEAR STRENGTH	ASTM D6243	500 psf (24 kPa) typ.@ 200 psf (9.6 kPa)

Bentomat® is a registered trademark of Cetco®.

¹ Geosynthetic property tests performed on the geosynthetic components before they are incorporated into the finished GCL product.

² Bentonite property tests performed before the bentonite is incorporated into the finished GCL product.

³ Reported at 0 percent moisture content.

⁴ Geofilm tensile break strength performed in the machine and cross-machine directions using ASTM D882.

⁵ GCL tensile strength testing is performed in the machine direction using ASTM D6768.

⁶ ASTM D5887 is modified to include the laminated thin flexible membrane on the test specimen. Index flux and hydraulic conductivity testing with deaired distilled/deionized water at 80 psi (550 kPa) cell pressure, 77 psi (530 kPa) headwater pressure and 75 psi (515 kPa) tailwater pressure. ASTM D5887 (modified) testing is performed only on a periodic basis because the thin flexible membrane is essentially impermeable. The Bentomat® GCL core (without the flexible membrane) has a maximum hydraulic conductivity of 5 x 10⁻¹¹ m/s with deaired distilled/deionized water. For more information, see CETCO® Technical Reference (TR) Nos. 111 and 112.

⁷ Peak values measured at 200 psf (9.6 kPa) normal stress for a specimen hydrated for 48 hours. Site-specific materials, GCL products, and test conditions must be used to verify internal and interface strength of the proposed design.

BENTOMAT®

Bentomat® CL is a reinforced composite laminate geosynthetic clay liner (GCL) consisting of bentonite clay and a flexible plastic membrane. The bentonite clay is encapsulated between two geotextiles that are needle-punched together, and the flexible plastic membrane is laminated to the outer geotextile surface. Composite laminate GCLs have a maximum permeability 200 times less than conventional compacted clay liners and 10 times less than standard bentonite-based GCLs. Bentonite needle-punched GCLs provide a durable reinforcement to withstand shear stresses on steep slopes and the high needle-punch density provides higher peak internal shear strengths.

Bentomat® CL geo clay liners can completely or partially replace thick, multi lift compacted clay layers in composite landfill liners and caps due to the efficiency of the high swelling natural sodium bentonite clay. Bentonite-based GCLs are designed to provide a hydraulic barrier against many leachates. The swelling bentonite fills up pore space and constricts flow paths to water, resulting in a low-permeability hydraulic barrier.

Note: To the best of our knowledge, unless otherwise stated, these are typical property values and are intended as guides only, not as specification limits. Chemical resistance, odor transmission, longevity as well as other performance criteria is not implied or given and actual testing must be performed for applicability in specific applications and/or conditions. RAVEN INDUSTRIES MAKES NO WARRANTIES AS TO THE FITNESS FOR A SPECIFIC USE OR MERCHANTABILITY OF PRODUCTS REFERRED TO, no guarantee of satisfactory results from reliance upon contained information or recommendations and disclaims all liability for resulting loss or damage.

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